

Master of Data Science for Public Policy

Sneak Peek: Introduction to Data Science

Simon Munzert
Hertie School

Welcome!

Introductions

Me

 I'm **Simon Munzert** [si'mən munsɜrt], or just Simon [saimən].

 munzert@hertie-school.org

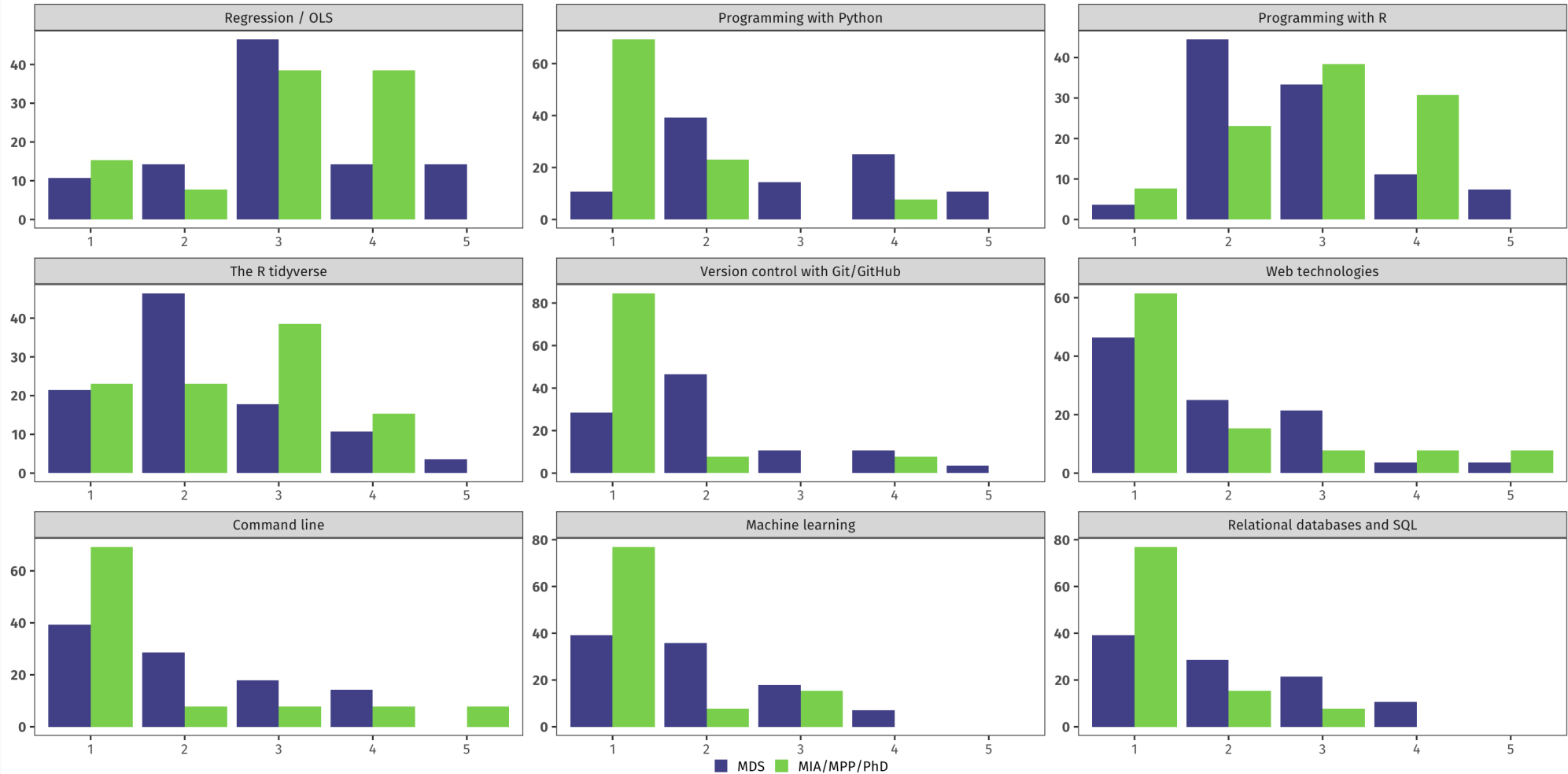
 Professor of Data Science and Public Policy | Director of the Data Science Lab

Introduction to Data Science

 <https://github.com/intro-to-data-science-25>

Much of this course lives on GitHub. You will find lecture materials, code, assignments, and other people's presentations there. We also have Moodle, which is for everything else.

More about "you"



The MDS in a nutshell

Curriculum

SEMESTER 1	✓ The Policy Process
	✓ Economics
	✓ Mathematics for data science
	✓ Introduction to data science
SEMESTER 2	✓ Causal inference
	✓ Machine learning
	✓ Data structures and algorithms
	✓ Law and governance
SEMESTER 3 + 4	✓ Internship
	✓ 2 Advanced tools electives
	✓ 2 Governance and Management electives
	✓ 2 Portfolio electives: Expand your policy and data science portfolio
	✓ Master's thesis

Curriculum

SEMESTER 1	✓ The Policy Process
	✓ Economics
	✓ Mathematics for data science
	✓ Introduction to data science
SEMESTER 2	✓ Causal inference
	✓ Machine learning
	✓ Data structures and algorithms
	✓ Law and governance
SEMESTER 3 + 4	✓ Internship
	✓ 2 Advanced tools electives
	✓ 2 Governance and Management electives
	✓ 2 Portfolio electives: Expand your policy and data science portfolio
	✓ Master's thesis

Prof. Asya Magazinnik, PhD | Professor of Social Data Science



I'm Asya Magazinnik and I am Professor of Social Data Science at the Hertie School. I study political institutions, with emphasis on electoral geography, federalism, local politics, and law enforcement and am also interested in political methodology. I teach Mathematics for data science in the first semester of the MDS, a course that will provide you with a broad understanding of linear algebra, probability theory, statistics and optimisation, all necessary for understanding the theoretical underpinnings of modern statistics and machine learning methods.

Curriculum

SEMESTER 1	✓ The Policy Process
	✓ Economics
	✓ Mathematics for data science
	✓ Introduction to data science
SEMESTER 2	✓ Causal inference
	✓ Machine learning
	✓ Data structures and algorithms
	✓ Law and governance
	✓ Internship
SEMESTER 3 + 4	✓ 2 Advanced tools electives
	✓ 2 Governance and Management electives
	✓ 2 Portfolio electives: Expand your policy and data science portfolio
	✓ Master's thesis

Simon Munzert, Professor of Data Science and Public Policy | Director, Data Science Lab



I'm Simon Munzert and I am Professor of Data Science and Public Policy and Director of the Data Science Lab at the Hertie School. I teach **Introduction to Data Science** in the first semester of the MDS. This course will get you up to speed with the modern data science workflow with R. We will cover topics like version control (Git) and project management, web data collection, database storage, data wrangling, debugging, automation, and data science ethics. I've written a [textbook on web data collection with R](#). In my research, I have used large-scale survey and tracking data as well as experimental methods to learn [how to make people use important health apps](#), or [how political propaganda affects people's vote choice](#).

Curriculum

SEMESTER 1	✓ The Policy Process
	✓ Economics
	✓ Mathematics for data science
	✓ Introduction to data science
SEMESTER 2	✓ Causal inference
	✓ Machine learning
	✓ Data structures and algorithms
	✓ Law and governance
SEMESTER 3 + 4	✓ Internship
	✓ 2 Advanced tools electives
	✓ 2 Governance and Management electives
	✓ 2 Portfolio electives: Expand your policy and data science portfolio
	✓ Master's thesis

William Lowe, Senior Research Scientist

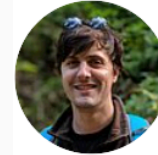


I'm William Lowe, a Senior Research Scientist at the Hertie School. My research covers legislative politics, political economy, and public policy, with a focus on statistical text analysis and causal inference. At Hertie, I teach **Causal Inference and Machine Learning**, where we look at the intersection of theoretical tools with practical applications in the analysis of data within the context of public policy, and **Data Science and Decision-Making**, where we study classical decision theory juxtaposed with the literature on human and machine cognitive performance and bias.

Curriculum

SEMESTER 1	✓ The Policy Process
	✓ Economics
	✓ Mathematics for data science
	✓ Introduction to data science
SEMESTER 2	✓ Causal inference
	✓ Machine learning
	✓ Data structures and algorithms
	✓ Law and governance
SEMESTER 3 + 4	✓ Internship
	✓ 2 Advanced tools electives
	✓ 2 Governance and Management electives
	✓ 2 Portfolio electives: Expand your policy and data science portfolio
	✓ Master's thesis

Drew Dimmery, Professor of Data Science for the Common Good



I'm Drew Dimmery and I am Professor of Data Science for the Common Good at the Hertie School. My research focuses on the intersection of machine learning and causal inference. I work a lot on experimental design and I think hard about how experimentation can be better and more efficient. I'm also interested in better applying the tools of machine learning to observational causal inference. I teach two courses in the second semester of the MDS. In [Data Structures & Algorithms](#), you will become capable software engineers able to design and analyze algorithms able to manage complex projects. In [Machine Learning](#), you will learn how and why machine learning works. In both classes you will get hands-on experience and projects you can add to your portfolio.

Curriculum

SEMESTER 1	✓ The Policy Process
	✓ Economics
	✓ Mathematics for data science
	✓ Introduction to data science
SEMESTER 2	✓ Causal inference
	✓ Machine learning
	✓ Data structures and algorithms
	✓ Law and governance
SEMESTER 3 + 4	✓ Internship
	✓ 2 Advanced tools electives
	✓ 2 Governance and Management electives
	✓ 2 Portfolio electives: Expand your policy and data science portfolio
	✓ Master's thesis

Prof. Lynn Kaack, PhD | Assistant Professor of Computer Science and Public Policy



I'm Lynn Kaack and I am Assistant Professor of Computer Science and Public Policy at the Hertie School. My research focuses on methods from statistics and machine learning to inform climate mitigation policy across. I'm also a co-founder and chair of [Climate Change AI](#), which works at the intersection of machine learning and climate action. I teach **Artificial intelligence and climate change** in the second year of the MDS, a course that will give you insights into cutting edge research using AI and machine learning to address climate change. I also teach **Deep Learning** in the second year of the MDS, where you can learn the main theoretical concepts of (deep) neural networks, and its applications in computer vision, natural language processing and other areas.

What is data science?

What is data science?

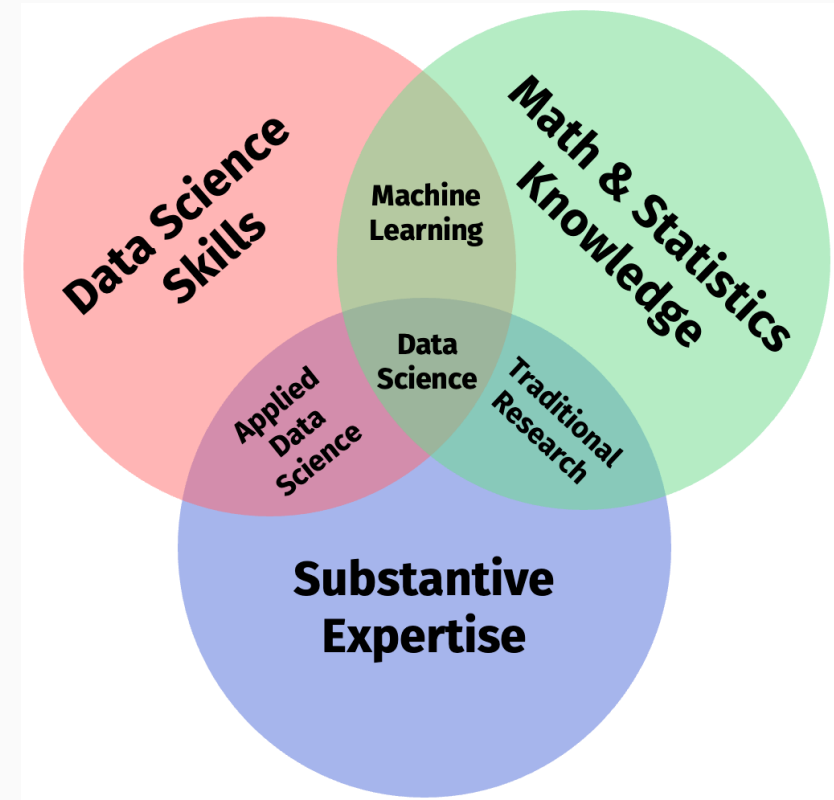
What is data science?

"Data science is an **interdisciplinary academic field** that uses statistics, scientific computing, scientific methods, processes, algorithms and systems to extract or extrapolate knowledge and insights from potentially noisy, structured, or unstructured data." - [Wikipedia](#)

"Data science is a concept to **unify statistics, data analysis, informatics**, and their related methods to understand and analyze actual phenomena with data." - [Chikio Hayashi](#)

"Data science encompasses a **set of principles**, problem definitions, algorithms, and processes for **extracting nonobvious and useful patterns from large data sets**." - [John Kelleher and Brendan Tierney](#)

A working definition



Source [Drew Conway, 2010](#) (adapted)

The data science pipeline



The data science pipeline

Preparatory work

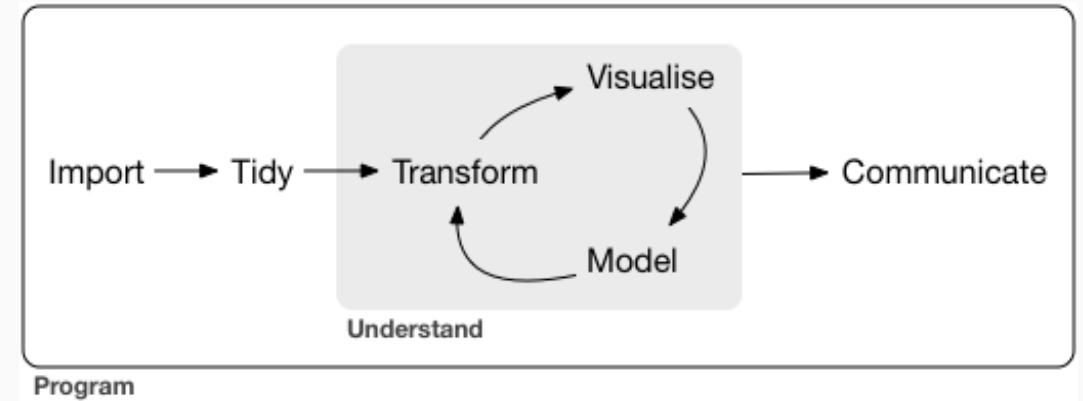
- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

The data science pipeline

Preparatory work

- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

Data operation



Source [H. Wickham, M. Çetinkaya-Rundel, G. Gronemund: R for Data Science](#)

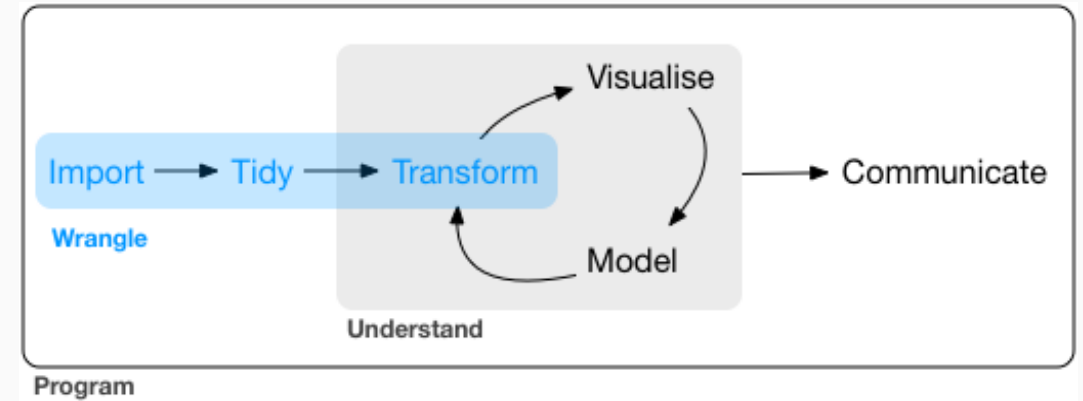
The data science pipeline

Preparatory work

- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

Data operation

- **Wrangle**: import, tidy, manipulate



Source H. Wickham, M. Çetinkaya-Rundel, G. Gronlund:
R for Data Science

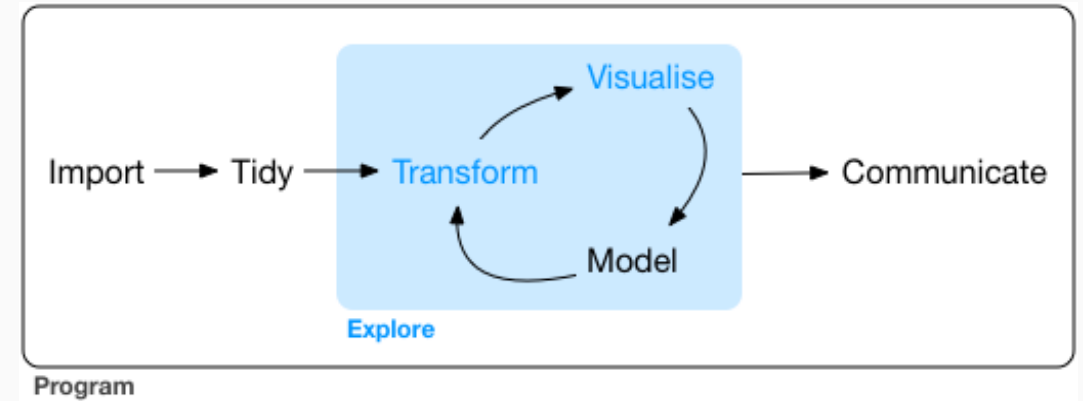
The data science pipeline

Preparatory work

- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

Data operation

- **Wrangle**: import, tidy, manipulate
- **Explore**: visualize, describe, discover



Source [H. Wickham, M. Çetinkaya-Rundel, G. Gronlund: R for Data Science](#)

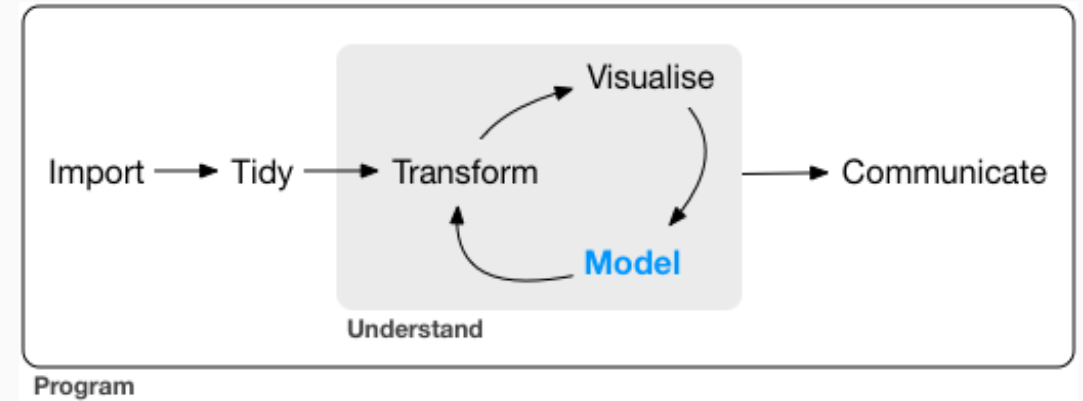
The data science pipeline

Preparatory work

- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

Data operation

- **Wrangle**: import, tidy, manipulate
- **Explore**: visualize, describe, discover
- **Model**: build, test, infer, predict



Source [H. Wickham, M. Çetinkaya-Rundel, G. Gronlund: R for Data Science](#)

The data science pipeline

Preparatory work

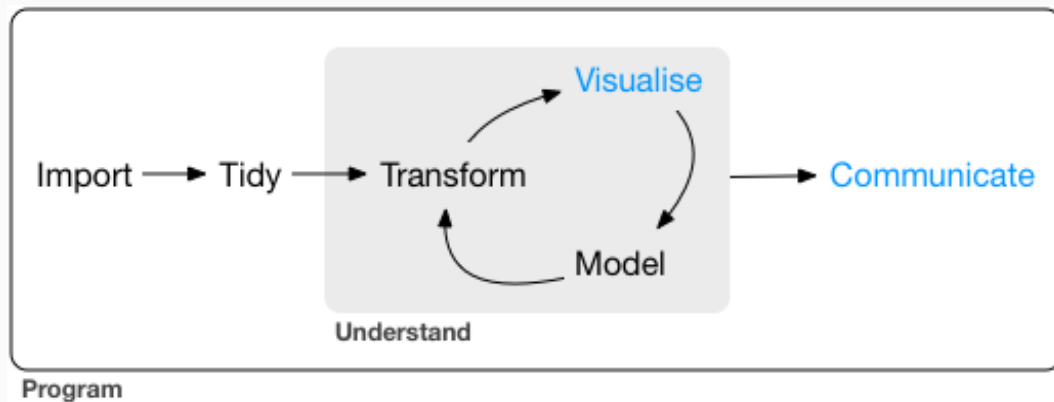
- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

Data operation

- **Wrangle**: import, tidy, manipulate
- **Explore**: visualize, describe, discover
- **Model**: build, test, infer, predict

Dissemination

- **Communicate**: to the public, media, policymakers
- **Publish**: journals/proceedings, blogs, software
- **Productize**: make usable, robust, scalable



Source [H. Wickham, M. Çetinkaya-Rundel, G. Gronlund: R for Data Science](#)

The data science pipeline

Preparatory work

- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

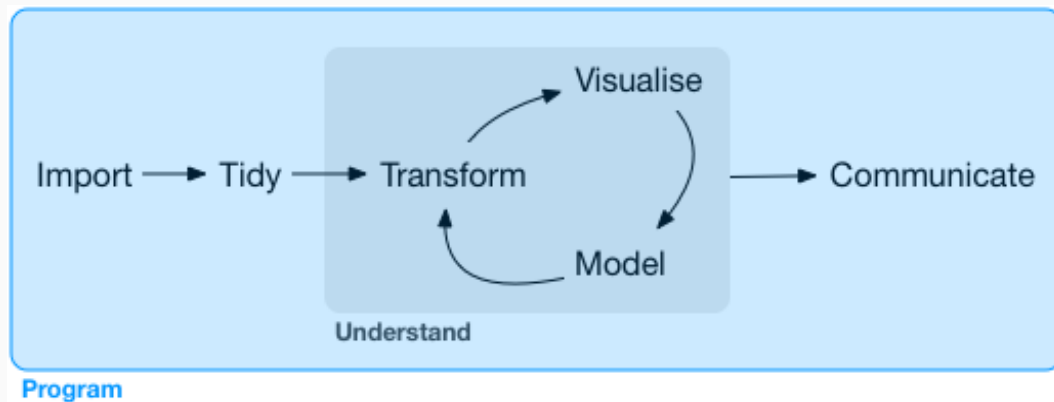
Data operation

- **Wrangle**: import, tidy, manipulate
- **Explore**: visualize, describe, discover
- **Model**: build, test, infer, predict

Dissemination

- **Communicate**: to the public, media, policymakers
- **Publish**: journals/proceedings, blogs, software
- **Productize**: make usable, robust, scalable

Meta skill: programming



Source [H. Wickham, M. Çetinkaya-Rundel, G. Gronlund: R for Data Science](#)

The data science pipeline

Preparatory work

- **Problem definition** predict, infer, describe
- **Design** conceptualize, build data collection device
- **Data collection** recruit, collect, monitor

Data operation

- **Wrangle**: import, tidy, manipulate
- **Explore**: visualize, describe, discover
- **Model**: build, test, infer, predict

Dissemination

- **Communicate**: to the public, media, policymakers
- **Publish**: journals/proceedings, blogs, software
- **Productize**: make usable, robust, scalable

Meta skill: programming with R (and Python!)



Sneak preview

Introduction to Data Science in a nutshell

Session	Session Title	A	Date
<u>Fundamentals</u>			
1	What is data science?	-	September 08
2	Programming I: Project management, coding etiquette, functions	H	September 15
3	Programming II: Iteration, automation, scheduling	Q	September 22
<u>Collecting data</u>			
4	Web data and technologies	Q	September 29 October 1
5	Web scraping and APIs	H	October 06
6	Relational databases and SQL	Q	October 13
<u>Analyzing data</u>			
7	Modeling	Q	October 20
Fall break: no classes			
8	Visualization	H	November 03
<u>Communication and big picture</u>			
9	Monitoring and communication	H	November 10
10	Data science ethics	Q	November 17
11	Towards open data science	-	November 24
12	Hackathon	-	December 01
Final Exam Week: no classes			

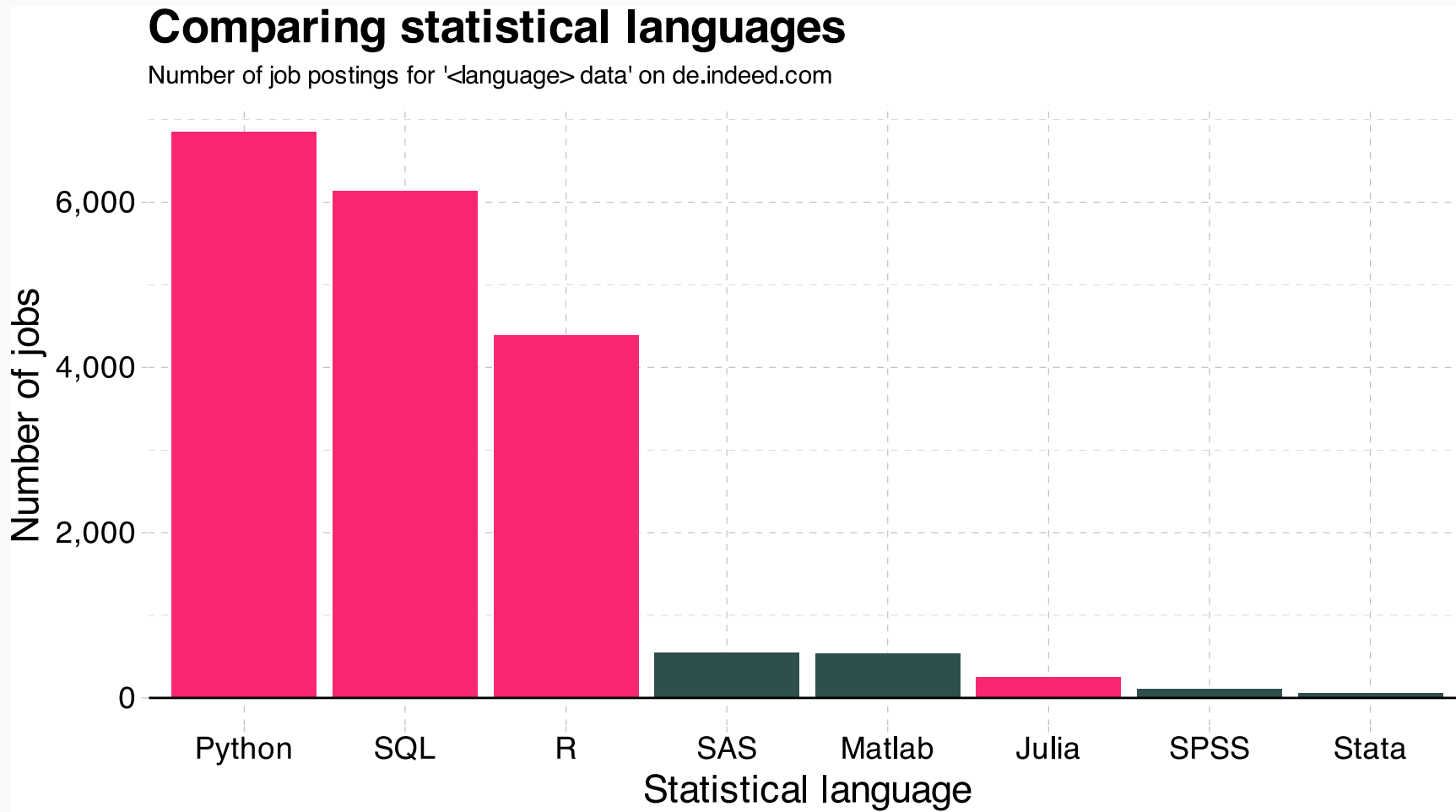
Sneak preview

Learning to love a programming environment

The tidyverse



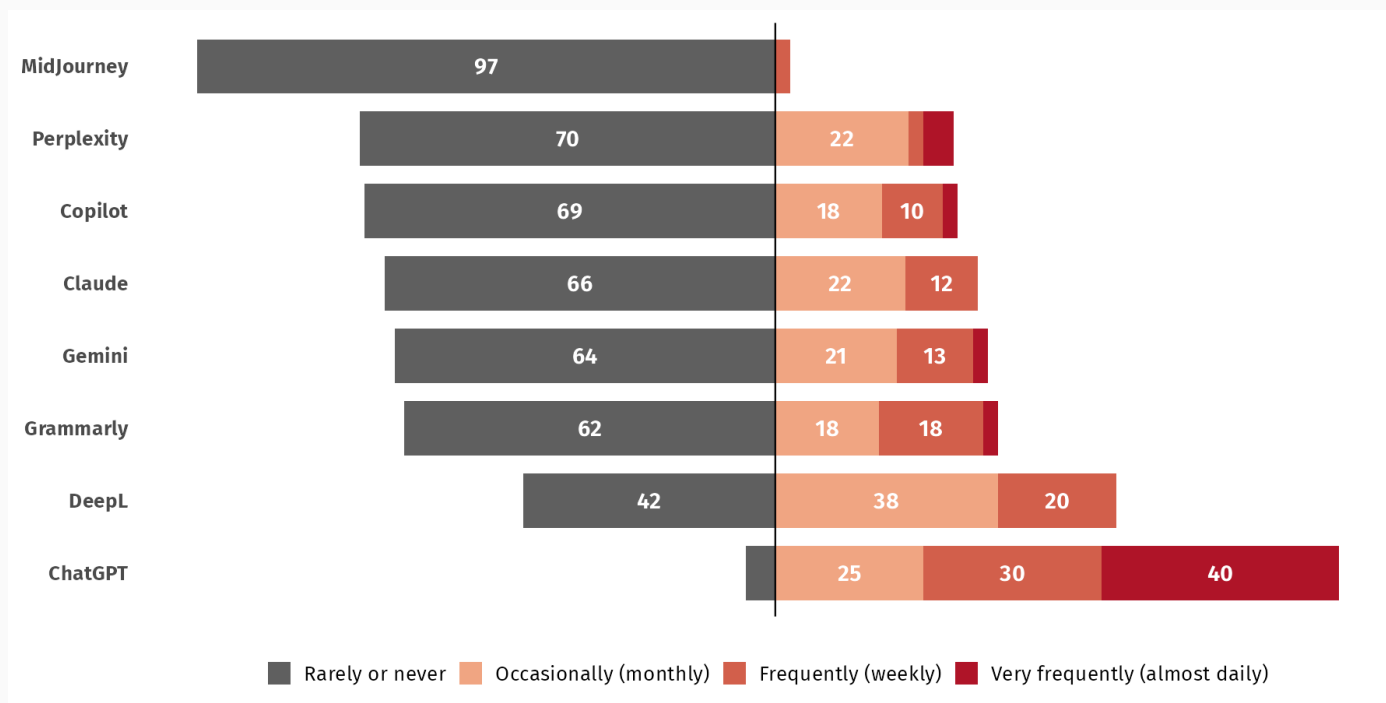
Why R and RStudio? (cont.)



AI use in and for the course

Can AI tools (LLM-based coding assistants) be used for assignments?

Yes, but use them with care. You will not become an efficient programmer if you heavily rely on those tools without learning the basics.



AI use in and for the course

Should you use AI tools for this course?

- You are a student at one of the leading policy schools in the world. You decided to invest a considerable amount of time and money to learn about data science and public policy here. I think you're at the right place to do so.
- However, the learning is still on you. What is key is that you want to *understand* what you are doing. That will require you to think hard and work hard. And that is ultimately what you need to excel in a data-based job.
- Or, you could just rely on some commercial language model to do the hard work for you. If you do that, you will notice that the muscles you wanted to train will not grow.
- That is not to say that AI is not useful for coding. It is incredibly powerful. But it also often fails spectacularly. You need to understand what is going on under the hood to make it work for you.
- I am not here to discipline and punish. I have no interest in that. What I can do is provide you with an environment that facilitates learning and sets the right incentives. If you want to learn, I will help you. If you don't, I can't.



Sneak preview

Collecting web data at scale

The MIT Billion Prices Project

Journal of Economic Perspectives—Volume 30, Number 2—Spring 2016—Pages 151–178

The Billion Prices Project: Using Online Prices for Measurement and Research

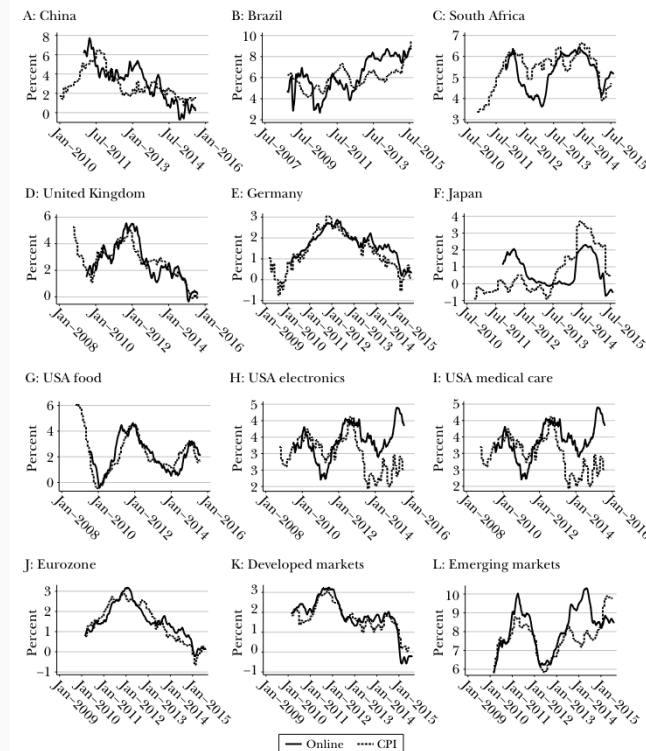
Alberto Cavallo and Roberto Rigobon

New data-gathering techniques, often referred to as “Big Data,” have the potential to improve statistics and empirical research in economics. This paper presents one example of how this can be achieved by using the vast number of online prices displayed on the web. We describe our work with the Billion Prices Project at MIT, and emphasize key lessons that can be used for both inflation measurement and some fundamental research questions in macro and international economics. In particular, we show how online prices can be used to construct daily price indexes in multiple countries and to avoid measurement biases that distort evidence of price stickiness and international relative prices.

The basic procedure used in most countries to collect inflation data has remained roughly the same for decades. A large number of people working for national statistical offices visit hundreds of stores on a monthly or bimonthly basis to collect prices for a preselected basket of goods and services. The micro data are then processed and used to construct consumer price indexes and other related indicators. This process is expensive, complex, and often too slow for some users of the data. Infrequent sampling and slow updates to the baskets can complicate

Figure 4

Online versus Consumer Price Index (CPI) Annual Inflation Rates



Source: Authors using online price indexes computed by PriceStats and consumer price indexes sourced from the national statistical office in each country.

Notes: Figure 4 compares inflation as measured by online prices and by the offline prices in the official consumer price index for a selection of countries, sectors, and regions. Annual inflation rates for daily online price indexes are computed as the percentage change in the average of the previous 30 days compared to the same average 365 days before. The series are nonseasonally adjusted. Indexes are “all-items” with the exception of China, where an online supermarket index is shown next to the official food index. Global aggregates in the last row are computed using 2010 consumption weights in each country and CPIs from official sources.

The MIT Billion Prices Project

Journal of Economic Perspectives—Volume 30, Number 2—Spring 2016—Pages 151–178

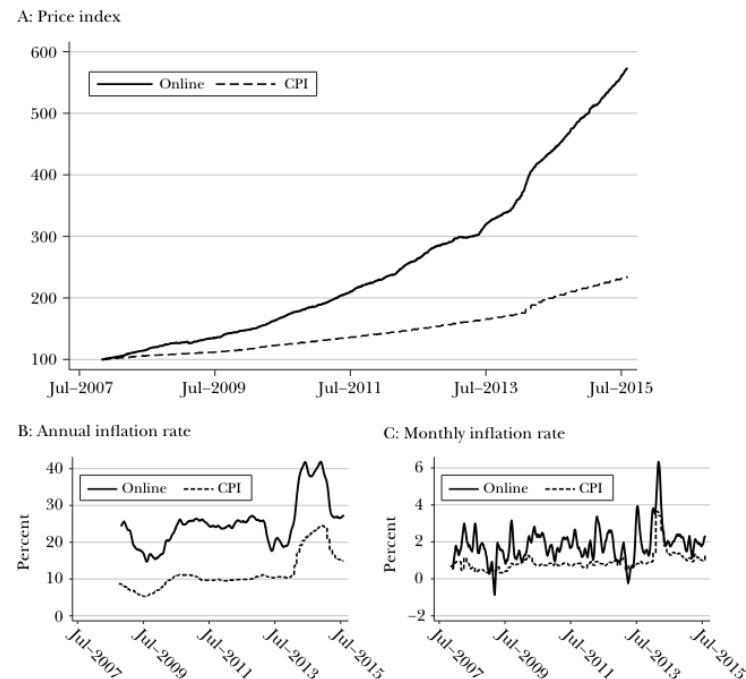
The Billion Prices Project: Using Online Prices for Measurement and Research

Alberto Cavallo and Roberto Rigobon

New data-gathering techniques, often referred to as “Big Data,” have the potential to improve statistics and empirical research in economics. This paper presents one example of how this can be achieved by using the vast number of online prices displayed on the web. We describe our work with the Billion Prices Project at MIT, and emphasize key lessons that can be used for both inflation measurement and some fundamental research questions in macro and international economics. In particular, we show how online prices can be used to construct daily price indexes in multiple countries and to avoid measurement biases that distort evidence of price stickiness and international relative prices.

The basic procedure used in most countries to collect inflation data has remained roughly the same for decades. A large number of people working for national statistical offices visit hundreds of stores on a monthly or bimonthly basis to collect prices for a preselected basket of goods and services. The micro data are then processed and used to construct consumer price indexes and other related indicators. This process is expensive, complex, and often too slow for some users of the data. Infrequent sampling and slow updates to the baskets can complicate

Figure 1
Argentina



Source: Authors using online price index computed by PriceStats and the consumer price index from the national statistical office in Argentina (INDEC).

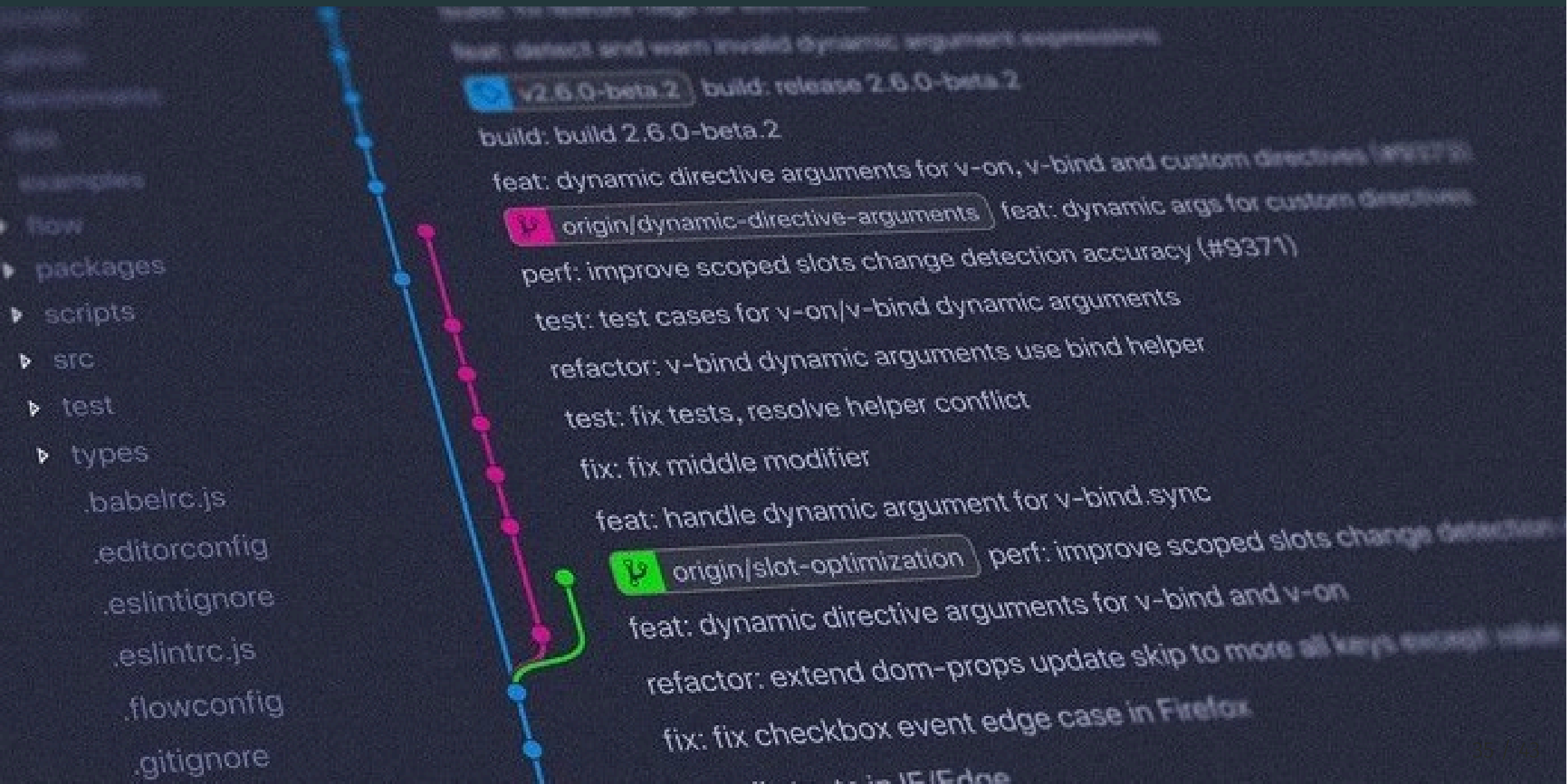
Notes: The figure compares a price index produced with online data to a comparable official consumer price index (CPI) for the case of Argentina from 2007 to 2015. It also looks at annual and monthly inflation rates using each source of data. Monthly inflation rates for the online index are computed as the percentage change in the average of the previous 30 days compared to the same average a month before. Annual inflation rates for the online index are computed as the percentage change in the average of the previous 30 days compared to the same average 365 days before. All price indexes are nonseasonally adjusted.

Sneak preview
Reflecting everyday ethics in data science

How do I respect intellectual property?



How do I ensure an open science workflow?



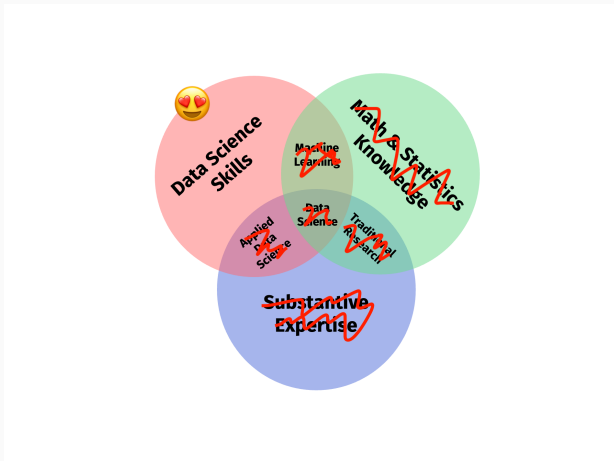
How do I communicate results honestly?



Summary

Goals of the course

- This course equips you with conceptual knowledge about the data science pipeline and coding workflow, data structures, and data wrangling.
- It enables you to apply this knowledge with statistical software.
- It prepares you for our other core courses and electives as well as the master's thesis.



What we will cover

- Version control and project management
- R and the tidyverse
- Programming workflow: debugging, automation, packaging
- Relational databases and SQL
- Web data and technologies
- Model fitting and evaluation
- Visualization
- Monitoring and communication
- Data science ethics
- (The command line)

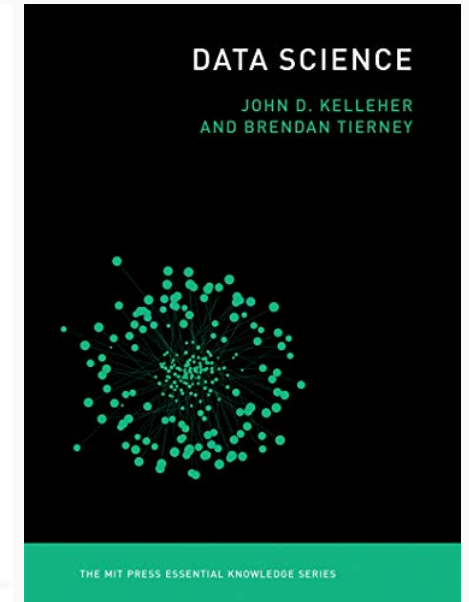
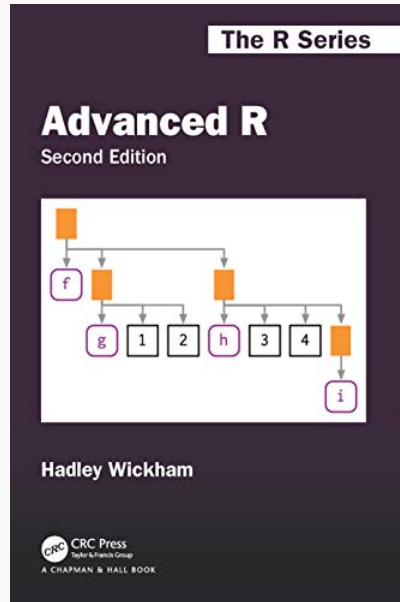
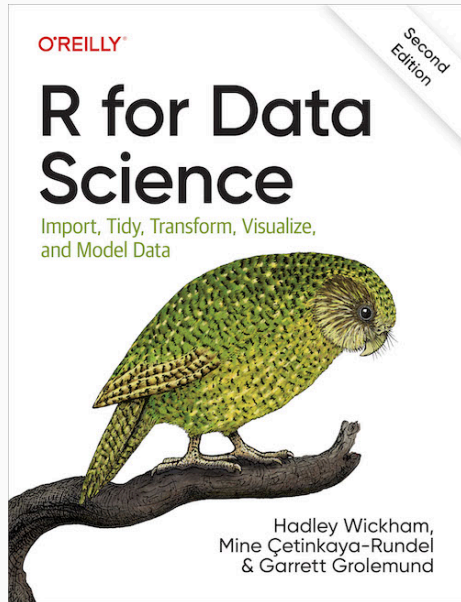
You at the beginning of the course



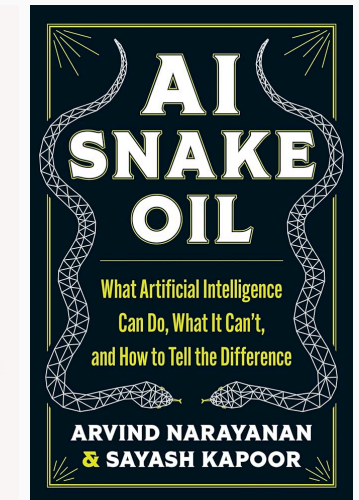
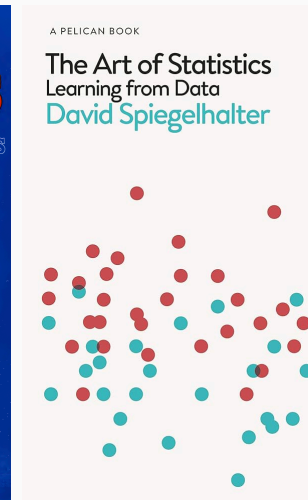
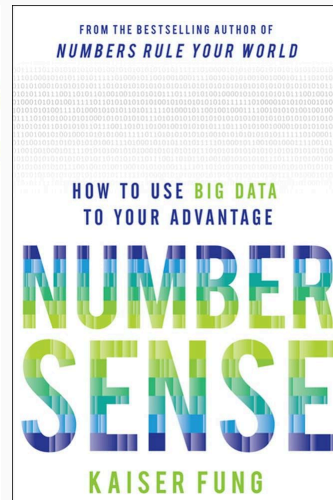
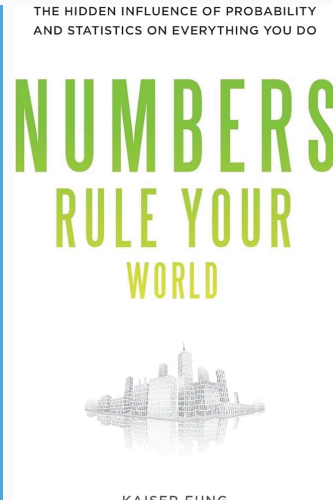
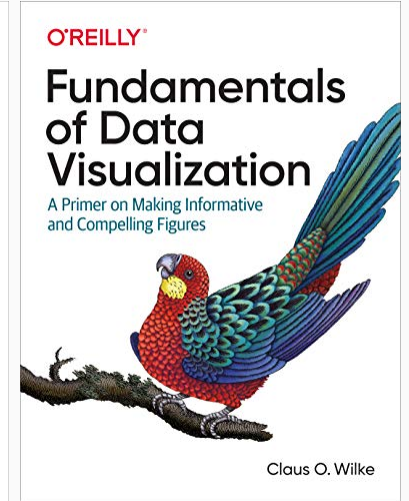
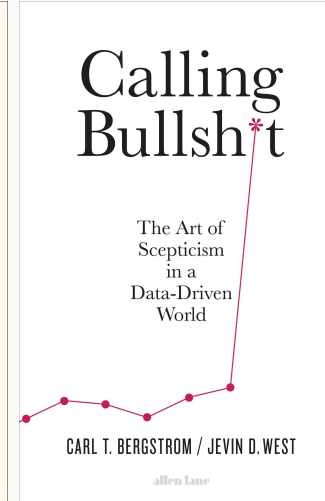
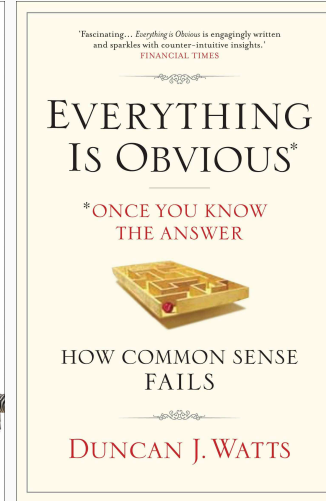
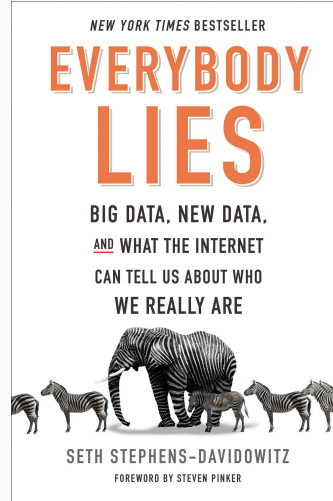
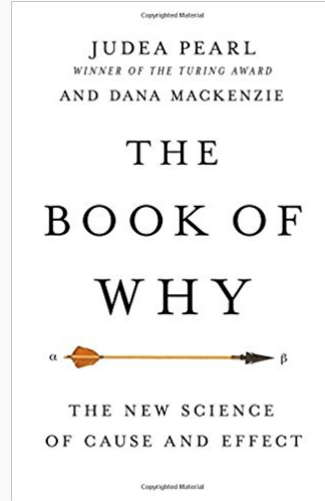
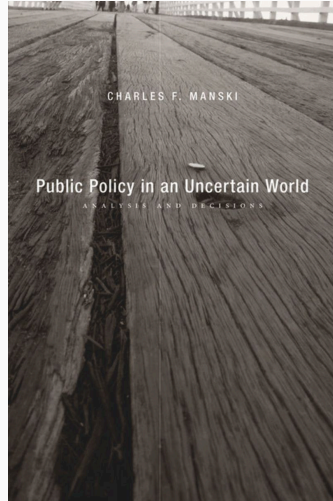
You at the end of the course



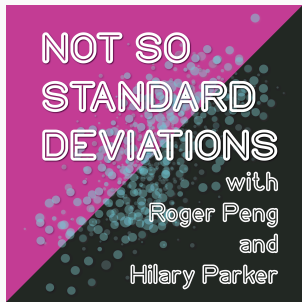
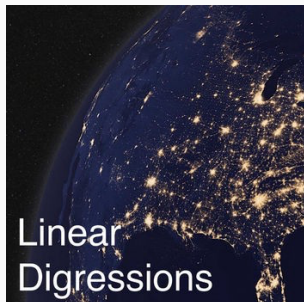
Core (and optional) readings



Further reading



Further listening



We look forward to welcoming you in Berlin in Fall!

